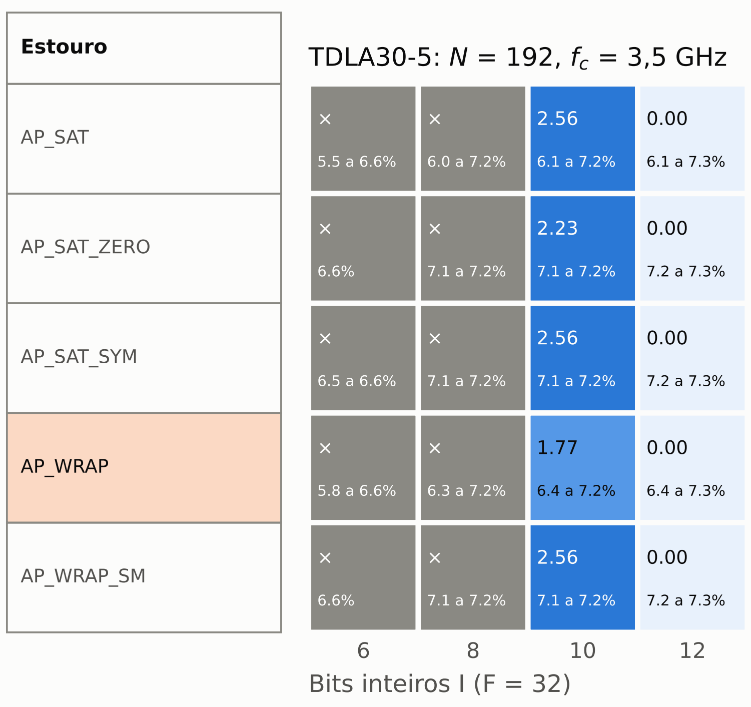
Configuração 12: $M = 256$, $C = 8$, $K = 12$, IoT = 20 dB, 16-QAM
Perda de SNR do BCD-MMSE (iter 4) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado (estouro: pela varredura de I; quantização: pela varredura de F), comparando de menos bits para mais, sem contar a coluna em que tudo é zero
Faixa "x a y": o outro modo muda o valor (a quantização no mapa de I, o estouro no mapa de F); se os dois extremos têm cores diferentes, a célula é dividida na diagonal (em cima, a cor do menor; embaixo, a do maior)
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo; sem número onde o formato e o modo não foram sintetizados

Perda de SNR (dB)

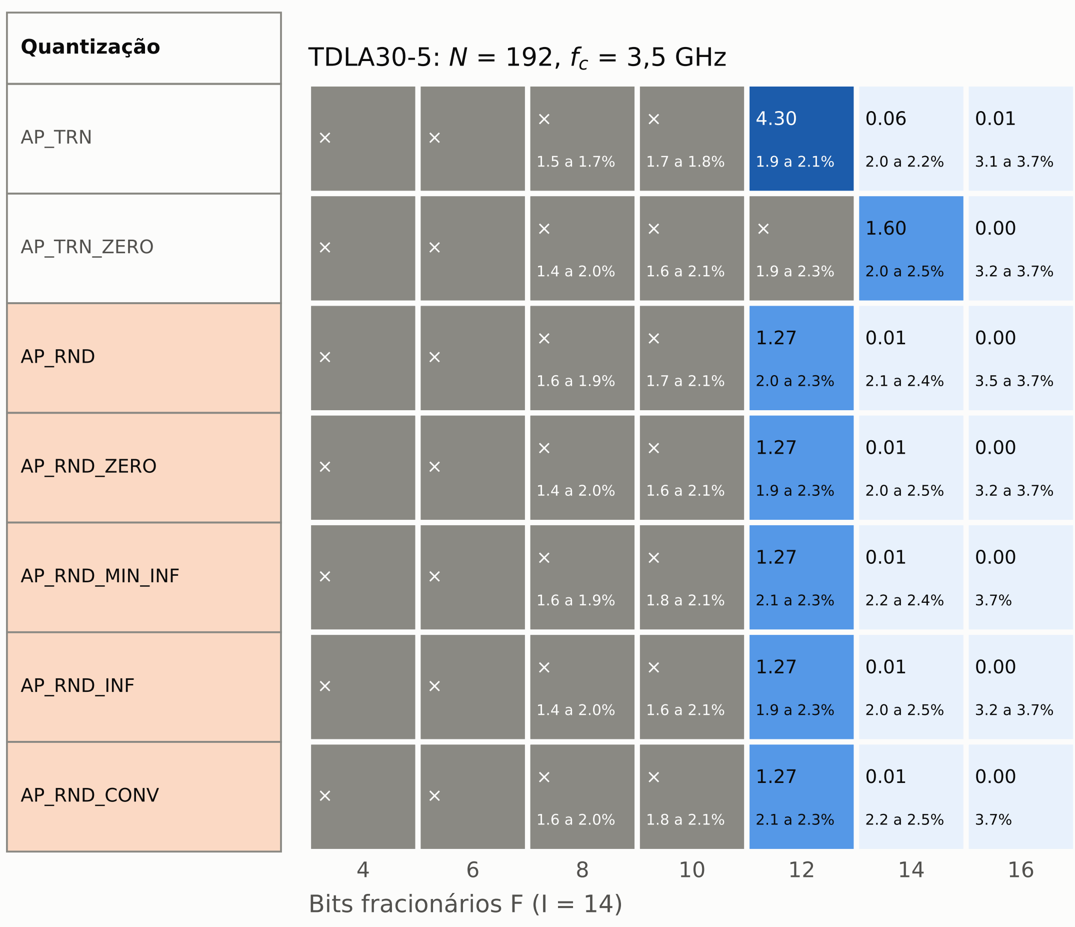
■ < 0,1 dB ■ 0,1 a 0,5 dB ■ 0,5 a 1 dB ■ 1 a 2 dB ■ 2 a 3 dB ■ 3 a 5 dB ■ 5 a 10 dB ■ > 10 dB ■ × não chega a 10^{-2} ■ rótulo do melhor modo



TDLA30-5: $N = 192$, $f_c = 3,5$ GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.56 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.23 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.56 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.77 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.56 7.1 a 7.2%	0.00 7.2 a 7.3%

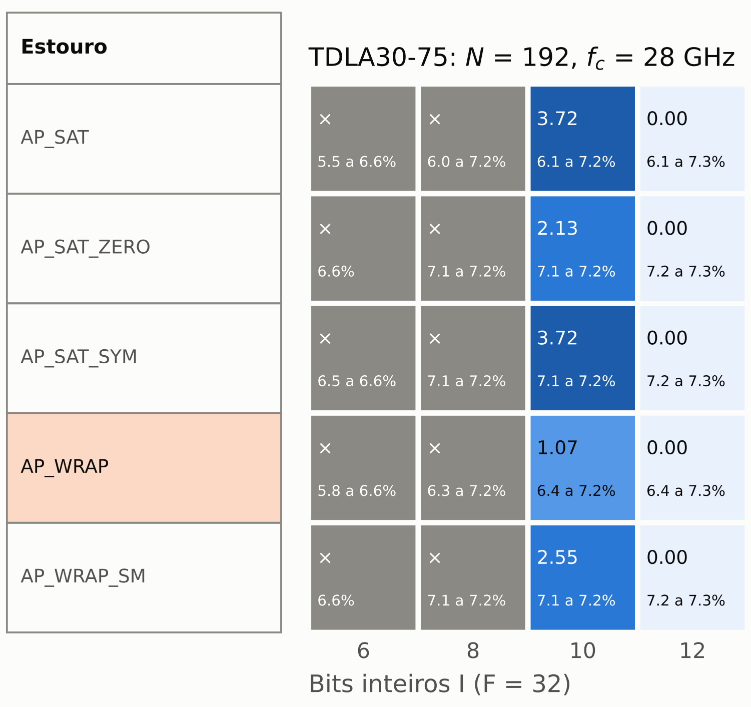
Bits inteiros I (F = 32)



TDLA30-5: $N = 192$, $f_c = 3,5$ GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	4.30 1.9 a 2.1%	0.06 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.60 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.27 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.27 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.27 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.27 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.27 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

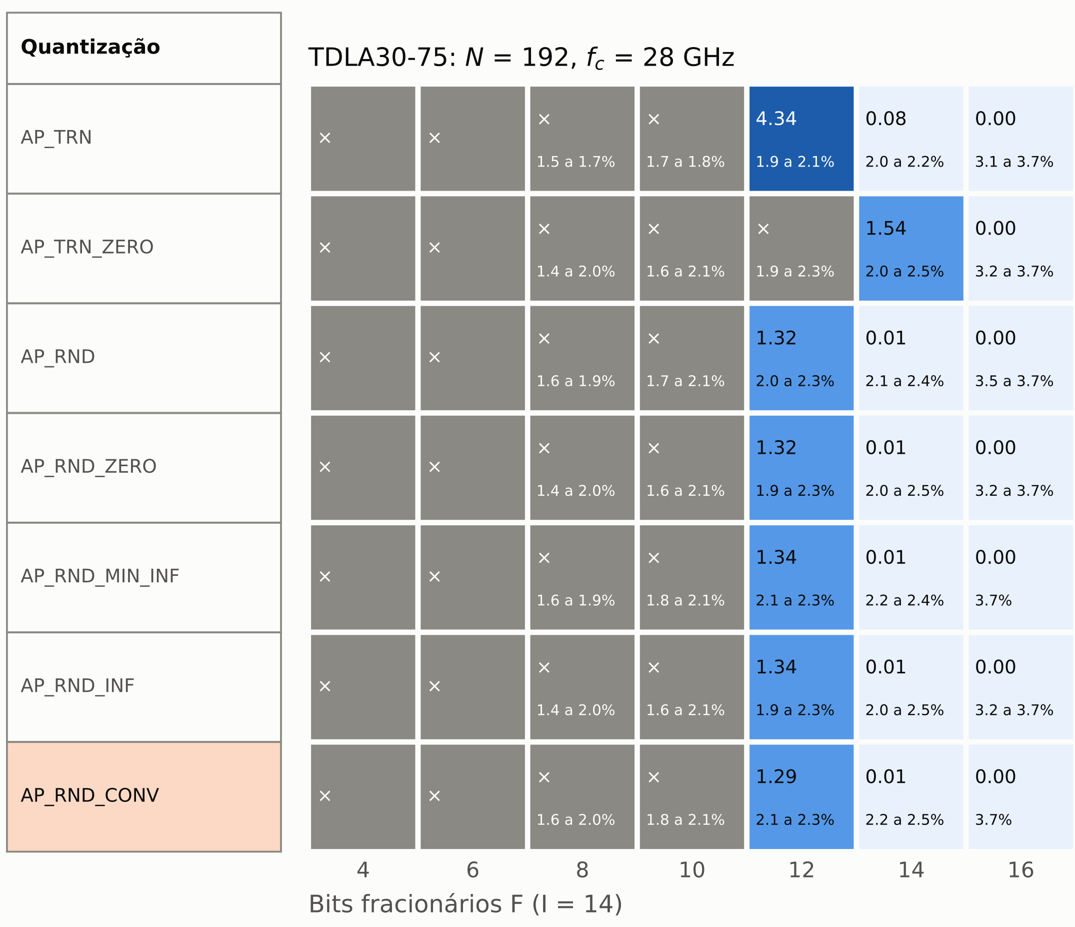
Bits fracionários F (I = 14)



TDLA30-75: $N = 192$, $f_c = 28$ GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	3.72 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.13 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	3.72 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.07 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.55 7.1 a 7.2%	0.00 7.2 a 7.3%

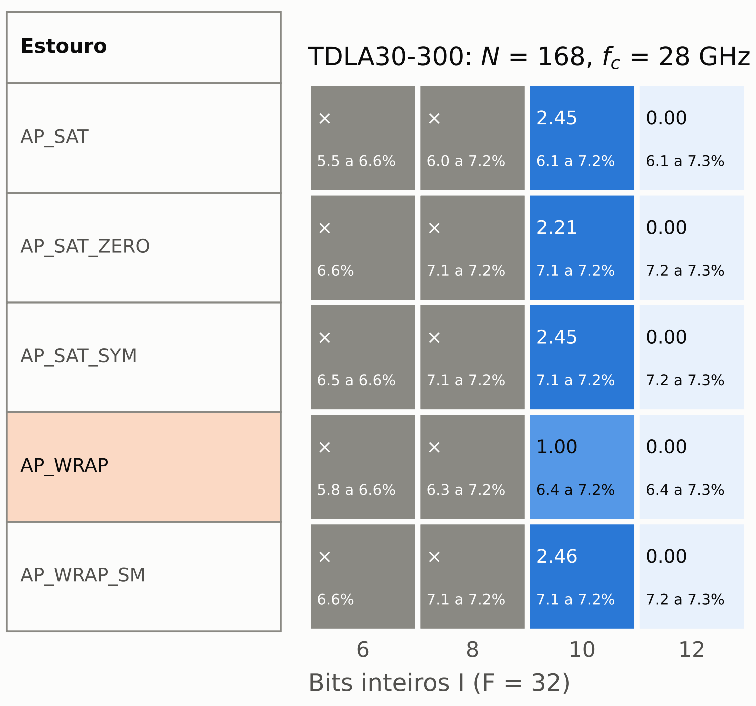
Bits inteiros I (F = 32)



TDLA30-75: $N = 192$, $f_c = 28$ GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	4.34 1.9 a 2.1%	0.08 2.0 a 2.2%	0.00 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.54 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.32 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.32 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.34 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.34 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.29 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

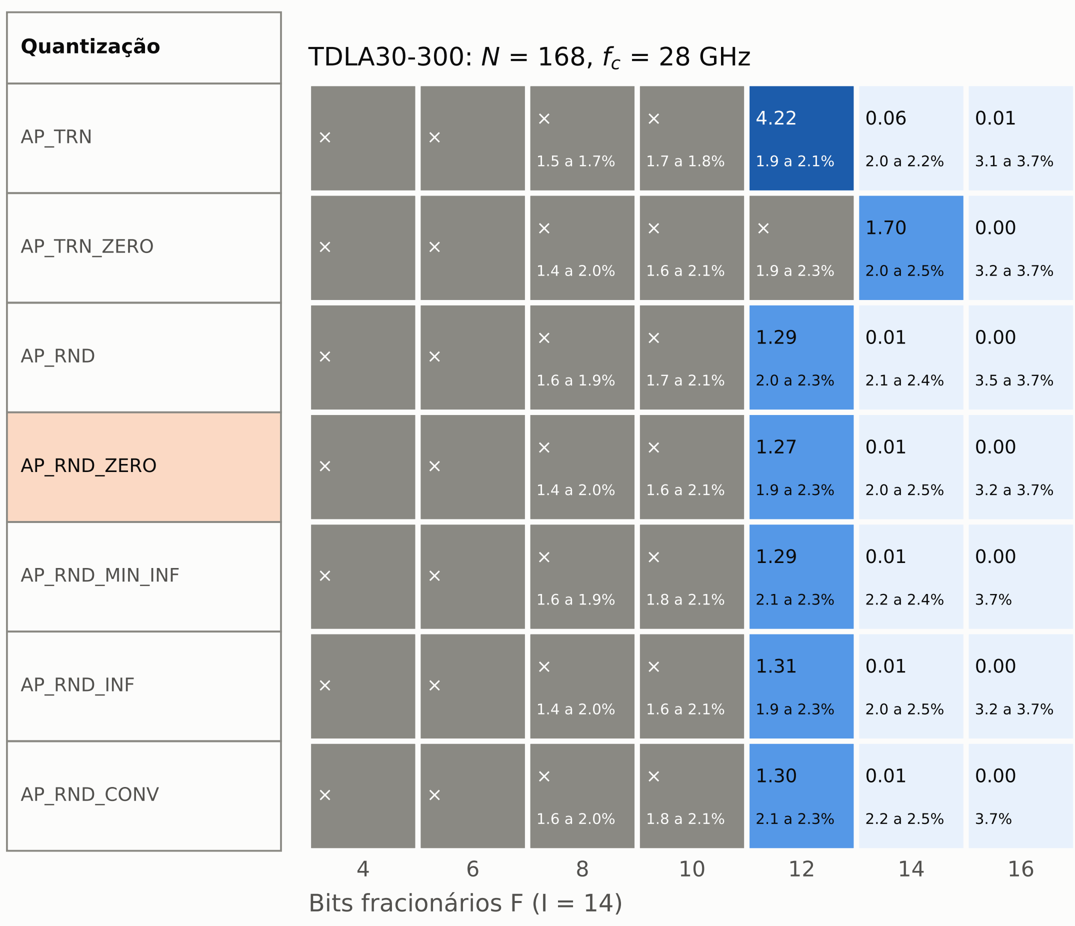
Bits fracionários F (I = 14)



TDLA30-300: $N = 168$, $f_c = 28$ GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.45 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.21 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.45 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.00 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.46 7.1 a 7.2%	0.00 7.2 a 7.3%

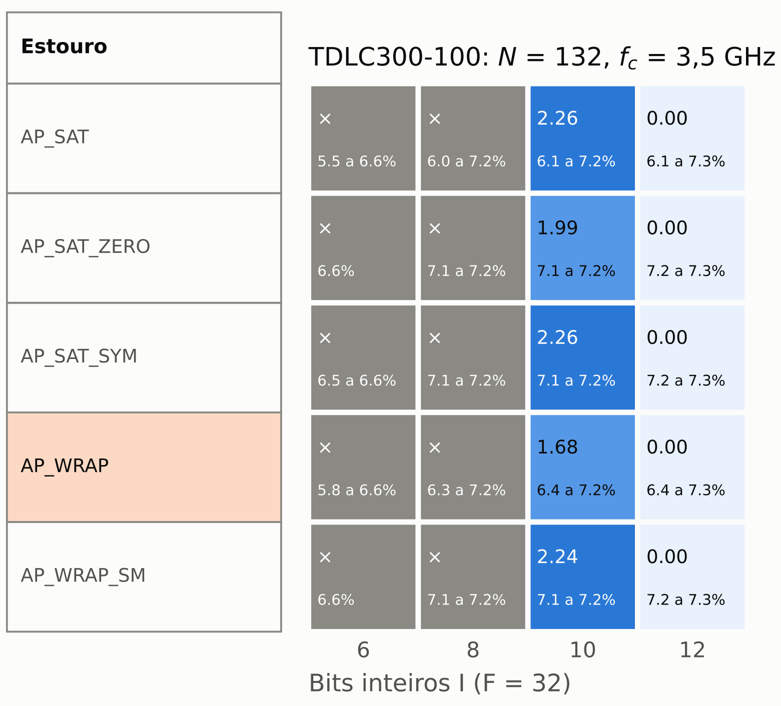
Bits inteiros I (F = 32)



TDLA30-300: $N = 168$, $f_c = 28$ GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	4.22 1.9 a 2.1%	0.06 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.70 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.29 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.27 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.29 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.31 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.30 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

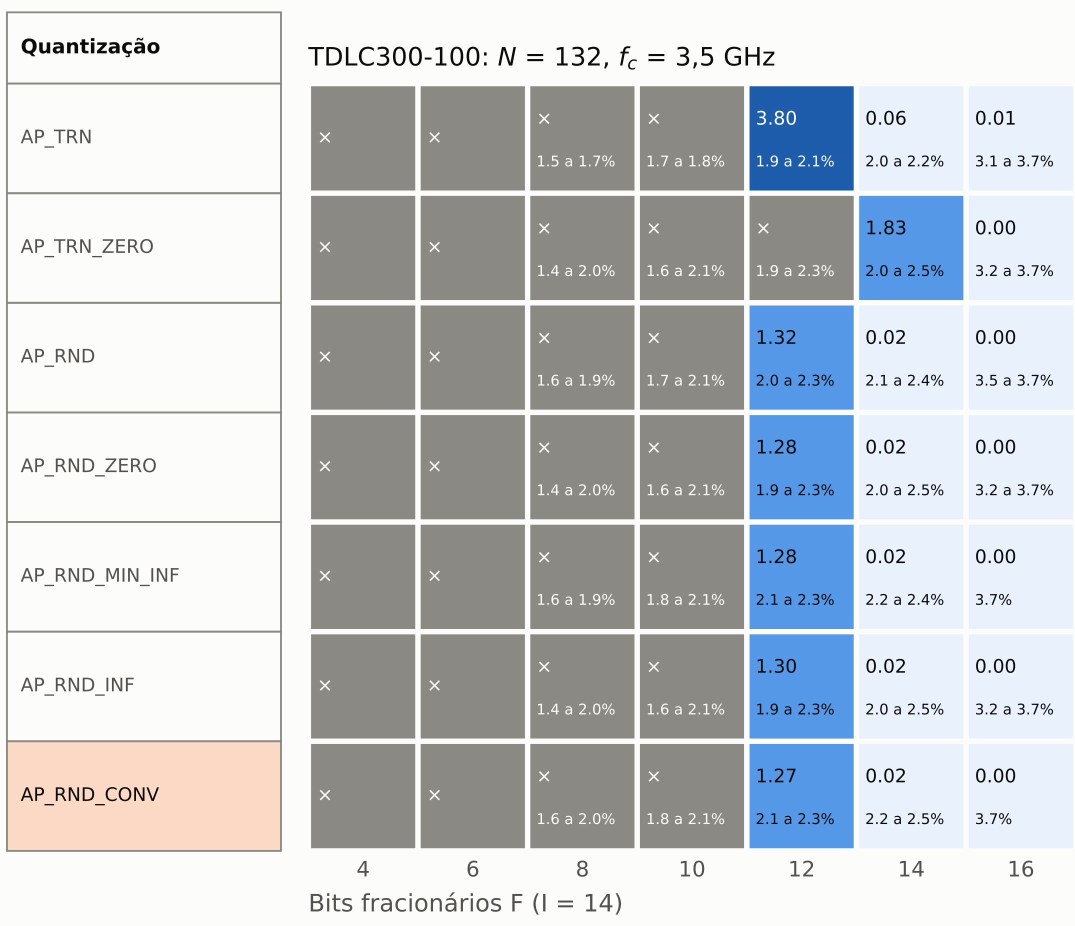
Bits fracionários F (I = 14)



TDLC300-100: $N = 132$, $f_c = 3,5$ GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.26 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	1.99 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.26 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.68 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.24 7.1 a 7.2%	0.00 7.2 a 7.3%

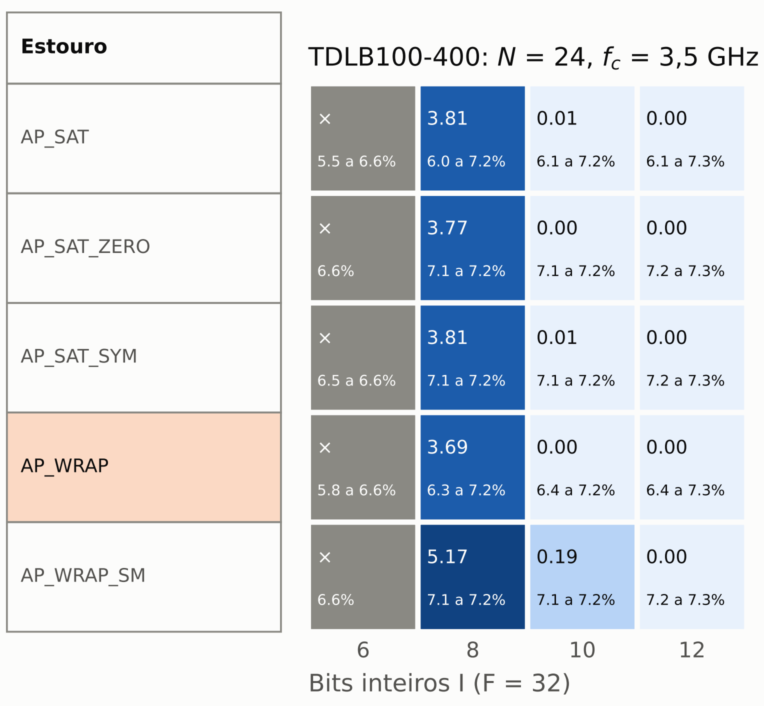
Bits inteiros I (F = 32)



TDLC300-100: $N = 132$, $f_c = 3,5$ GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	3.80 1.9 a 2.1%	0.06 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.83 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.32 2.0 a 2.3%	0.02 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.28 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.28 2.1 a 2.3%	0.02 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.30 1.9 a 2.3%	0.02 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.27 2.1 a 2.3%	0.02 2.2 a 2.5%	0.00 3.7%

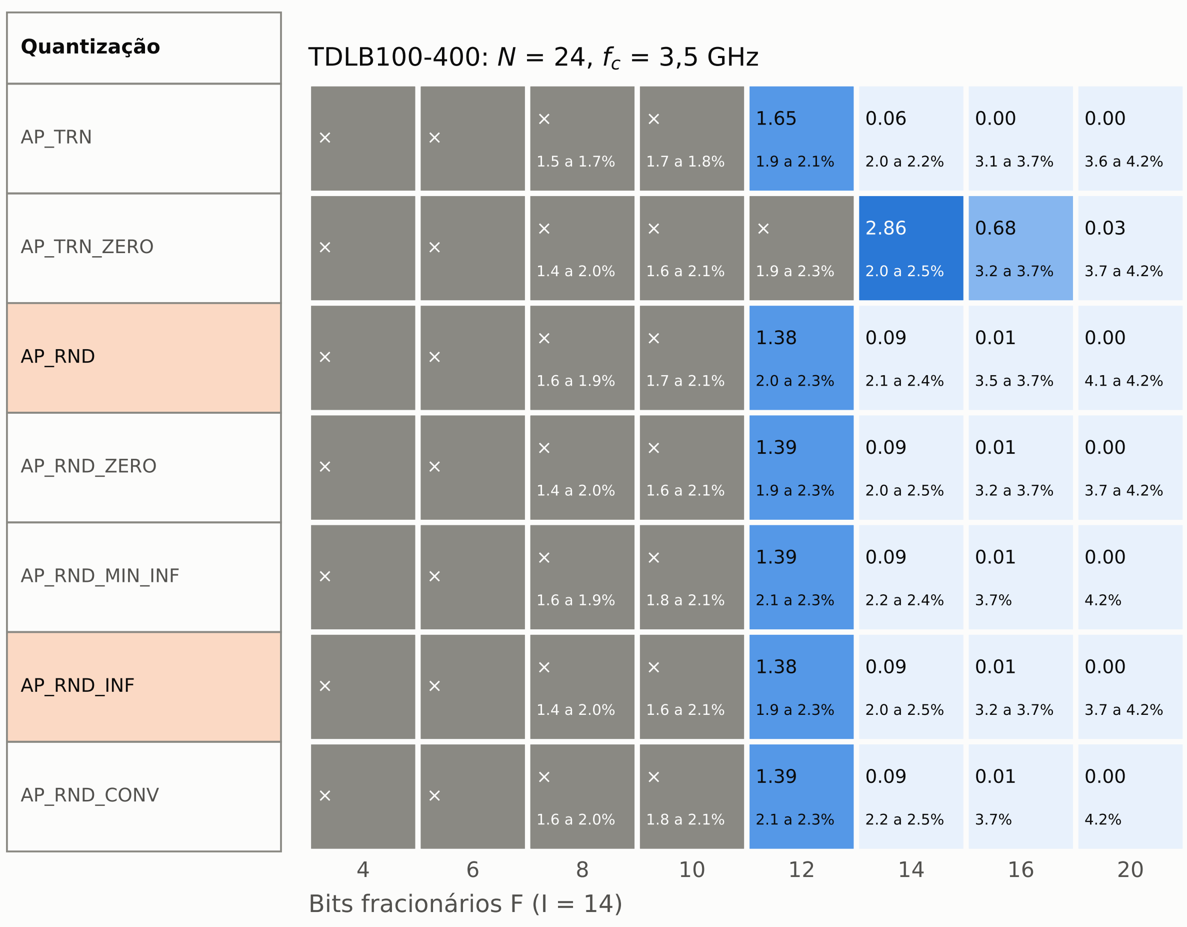
Bits fracionários F (I = 14)



TDLB100-400: $N = 24$, $f_c = 3,5$ GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	3.81 6.0 a 7.2%	0.01 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	3.77 7.1 a 7.2%	0.00 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	3.81 7.1 a 7.2%	0.01 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	3.69 6.3 a 7.2%	0.00 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	5.17 7.1 a 7.2%	0.19 7.1 a 7.2%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)



TDLB100-400: $N = 24$, $f_c = 3,5$ GHz

Quantização	4	6	8	10	12	14	16	20
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.65 1.9 a 2.1%	0.06 2.0 a 2.2%	0.00 3.1 a 3.7%	0.00 3.6 a 4.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	2.86 2.0 a 2.5%	0.68 3.2 a 3.7%	0.03 3.7 a 4.2%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.38 2.0 a 2.3%	0.09 2.1 a 2.4%	0.01 3.5 a 3.7%	0.00 4.1 a 4.2%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.39 1.9 a 2.3%	0.09 2.0 a 2.5%	0.01 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.39 2.1 a 2.3%	0.09 2.2 a 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.38 1.9 a 2.3%	0.09 2.0 a 2.5%	0.01 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.39 2.1 a 2.3%	0.09 2.2 a 2.5%	0.01 3.7%	0.00 4.2%

Bits fracionários F (I = 14)